MIL-C-81395A(AS)
8 December 1981
SUPERSEDING
MIL-C-81395(AS)
30 August 1966

MILITARY SPECIFICATION

CLOTH, COATED, DUCK, NYLON

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of nylon base natural or synthetic natural rubber coated cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging and Packing of Coated (Plastic, Rubber) and Laminated Fabrics.

MILITARY

MIL-C-7219 - Cloth, Duck, Nylon.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.
FED-STD-601 - Rubber, Sampling and Testing.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1487 - Glossary of Cloth Coating Imperfections.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of the specification shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. The guide sample, when furnished, shall be used solely for guidance and information to the contractor (see 6.3). If variation from this specification appears in the guide sample, this specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish one sample unit for first article inspection and approval (see 4 3 and 6.4).

3.3 Material

3.3.1 Base cloth. The base cloth shall conform to type III of MIL-C-7219, except that air permeability and water resistance shall not apply. The color shall be sage green, Air Force shade no. 1535.

3.3.2 Coating compound. The base cloth shall be coated on one side with 2.00 ± 0.25 ounces natural or synthetic natural rubber. The coating compound shall contain not less than 80 percent (by volume) of new plantation rubber or synthetic natural rubber pigmented black, plus softeners, curing agents, anti-oxidants and reinforcing materials prepared in such proportions as to meet the requirements specified in table I, when cured in sheet form to the same degree of cure and in the same manner as the finished coated cloth. The compound used in the outer coatings shall be pigmented during the compounding process so that the cured coating shall have a uniform and permanent color as specified in 3.5. The pigment shall be of such fineness that 100 percent of the pigment shall pass through a standard 325 mesh screen. The foundation coating and rubber coating compounds shall not be injurious to the base cloth and shall contain no compounding ingredients which would bloom to the surface or which would adversely affect the specified properties of the finished cloth. The compounding ingredients shall be prepared in such proportions as to cure properly and provide proofing films which shall be insoluble in water. When cured, the coating compound shall conform to the requirements specified in table I when tested as specified in table III.

3.4 Coated cloth. The base cloth shall be coated on one side only with the coating compound specified in 3.3.2. The coated cloth shall conform to the requirements specified in table II when tested as specified in table VI.

3.5 Color. The color of the finished cloth shall be black on the coated side and sage green, Air Force shade no. 1535 on the uncoated side (see 6.2).

3.6 Width. The width (plus 1 or minus 0 inch) of the coated cloth shall be as specified by the contracting officer (see 6.2).

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3.7 Length and put-up. The coated cloth shall be put-up in rolls as specified in PPP-P-1136. Each roll shall contain from 80 to 120 yards with no more than 3 pieces and no piece shall be less than 20 yards in length.

3.8 Workmanship. The finished coated cloth shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. The certificate of compliance, when certificates of compliance are submitted, shall certify that the coating compound conforms to the requirements specified in 3.3.2. The Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.2.1 Inspection conditions. Unless otherwise specified, all tests shall be performed in accordance with the test conditions specified in FED-STD-191 or the applicable test method referenced in the specification.

4.3 First article inspection. The first article submitted in accordance with 3.2 shall be visually examined in accordance with 4.4.2 and tested in accordance with the methods specified in 4.6. for the requirements specified in tables III and VI. The sample for testing shall consist of 3 yards, full width, of the coated cloth and 2 square feet (less than 0.125 inch in thickness) of the cured coating compound.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification. The coating compound shall be tested for the characteristics specified in table III. Unless otherwise specified, the testing shall be in accordance with the conditions and referenced test methods of FED-STD-601. The sample shall be two 12 by 12-inch specimens of the cured coating compound. The specimens shall not exceed 1/8-inch in thickness. The required tests shall be performed on each specimen. The lot shall be unacceptable if either specimen fails to meet any requirement specified.

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4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1, 4.4.2.2 and 4.4.2.3.

4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be inspected on the coated side only. Visual defects shall be as listed in table IV. The same yardage shall be given a through-light inspection for pinholes and lightly coated areas. The through-lighting inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for the examination shall be one linear yard. The acceptable quality level (AQL) for this examination, including the pinholes and lightly coated areas found in the through-light examination shall be 2.5 major and 6.5 total defects per 100 units (yards). The inspection level shall be level II of MIL-STD-105. The number of rolls from which the samples are to be selected shall be in accordance with table V. An approximately equal number of yards shall be examined in each roll sampled.

4.4.2.2 Overall examination. Each defect listed below shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number (maximum number of defects acceptable) shall be as shown in table V. The rolls selected for the yard-by-yard examination shall be included in this examination.

Overall defects

Selvage curled, folded, rolled or slack continuously or intermittently throughout the piece.
Width less than specified.
Overall uncleanness or objectionable odor.
Uneven coating throughout the piece.
Off-shade, shaded end to end, side to side, side to center or throughout the piece.
Mottled, cloudy or streaky throughout the piece.
Date of coating application missing from end of roll or identification tag.

4.4.2.3 Examination for length.

4.4 2 3.1 Individual rolls. The rolls shall be examined for gross length. Any gross length found to be less than the specified minimum length, or any gross length found to be more than 2 yards less than the gross length marked on the roll ticket, shall be considered as a defect with respect to length. The sample size (number of rolls selected as sample) and the acceptance number (maximum number of defects acceptable) shall be as shown in table V. The rolls selected for the yard-by-yard examination shall be included in this examination. Any roll containing more than three pieces or any piece less than 20-yards in length shall be considered a defect with respect to length.

4.4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the tickets.

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4.5 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1136 to determine that packaging, packing and marking comply with Section 5 requirements of this specification.

4.6 Testing of the end item. Prior to testing, the cloth and test apparatus shall be preconditioned in accordance with the requirements of FED-STD-191. The applicable methods of testing specified in FED-STD-191 as listed in table VI shall be followed. The physical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test data shall consist of the individual values used in expressing the final result. The sample unit for testing shall be 3 yards, full width, of the finished cloth. The lot size shall be expressed in units of one yard each. The lot shall be unacceptable if one or more units fail to meet any test requirements specified. The sample size (number of sample units to be tested) shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

5. PACKAGING

5.1 Packaging, packing, and marking. Cloth shall be packaged, packed, and marked in accordance with the requirements of PPP-P-1136, except that, prior to rolling, the face of the coated cloth shall be dusted with a suitable talc to prevent tackiness and sticking. The levels of packaging and packing shall be specified by the contracting activity (see 6.2). In addition to the markings required, each tag shall be marked with the following information: "Date of Application of Coating." In addition, the following precautionary marking shall appear on all containers: "STORE IN A COOL DRY PLACE".

6. NOTES

6.1 Intended use. The coated cloth is intended for use in the fabrication of the LR-1 life raft carrying case and other applications.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When guide sample is available (see 3.1 and 6.3).
c. When first article is required (see 3.2, 4.3 and 6.4).
d. Color required (see 3.5).
e. Width required (see 3.6).
f. Quantity required.
g. Selection of the applicable level of packaging and packing (see 5.1 and 5.2).
h. Additional marking if required.

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirement identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
4.1.1	Certificate of compliance	DI-E-2121	----
4.3	First article inspection report	DI-4902	----

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Guide sample. For guide sample, contact the acquiring activity issuing the invitation for bids.

6.4 First article. When a first article is required, the item shall be inspected and tested (see 3.2 and 4.3). The first article should be a preproduction sample consisting of 3 yards, full width of the finished cloth and 2 square feet (less than 0.125 inch in thickness) of the cured coating compound. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS

(Project No. 8305-N723)

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TABLE I. Physical requirements of the cured coating compound.

Characteristics	Requirements
Composition	1/
Water extract, percent, maximum	1.0
Tensile strength:	
Original, psi, minimum	2300
After heat aging, percent loss, maximum	30
Elongation:	
Original, percent, minimum	450
After heat aging, percent loss, maximum	15
Shore hardness reading:	
Original	50 ± 10
Change after heat aging	± 5

1/ A contractor's certificate of compliance in accordance with 4.1.1 shall be acceptable for this characteristic.

TABLE II. Physical requirements of coated cloth.

Characteristics	Requirements
Weight, oz/sq yd, maximum	9.5
Breaking strength, pounds, minimum, ravel strip:	
Warp	325
Filling	275
Water resistance, low range hydrostatic pressure, cm of water, minimum	25
Adhesion of coating per inch width, lbs, minimum	3.0
Blocking, maximum, at 93° ± 1°C (200° ± 2°F)	2

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TABLE III. Testing of cured coating compound.

Characteristics	Requirement	FED-STD-601 test method
Water extract	Table I	6621
Tensile strength:	Table I	
Original		4111
After heat aging at 168 hrs at 100° ± 3°C		7221
Elongation:	Table I	
Original		4121
After heat aging at 168 hrs at 100 ± 3°C		7221
Shore hardness:	Table I	
Original		3021
After heat aging at 168 hrs at 100 ± 3°C		7221

TABLE IV. Classification of defects for coating.

Defects	Classification	
	Major	Minor
Blisters or lumps:		
- 1/4-square inch or less in combined directions, clearly visible 1/		X
- Over 1/4-square inch in combined directions, clearly visible 1/	X	
Coating faults:		
- Blooming	X	
- Creases	X	
- Folds - winding on tubes		X
- Uneven coating		
-Heavily coated areas, clearly visible 1/		X
- Thin areas where coating compound is insufficient	X	
Coating missing	X	
Cuts	X	
Pinholes and windows	X	
Foreign matter incorporated in coating	X	
Splice mark clearly visible on coated side: 1/		
- If coating is not removed		X
- If coating is missing	X	
Stains in coating surface:		
- Any spot or stain (compound, oil, dirt) over 1 inch in length in any direction, clearly visible 1/		X
Streaks:		
- Shade streaks in coating, clearly visible 1/		X
Tears	X	

1/ The term "clearly visible" shall be interpreted as meaning clearly visible at normal examination distance (approximately 3 feet).

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TABLE V. Sample size.

Lot size in yards	Sample size in rolls	Maximum number of defects acceptable
Up to 1200 1/	3	0
1201 to 3200	5	0
3201 to 10,000	8	0
10,001 to 35,000	13	0
35,001 to 150,000	20	1
150,001 and over	32	2

1/ If lot is fewer than three rolls, each roll in the lot shall be examined.

TABLE VI. Quality conformance tests.

Characteristics	Requirement	FED-STD-191 test method
Weight	Table II	5041
Breaking strength: Warp Filling	Table II	5104
Water resistance	Table II	5514
Adhesion of coating	Table II	5970
Blocking	Table II	5872

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark o*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER (*Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	. DATE OF SUBMISSION *(YYMM*

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